PHYSICS

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

31. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : The property of body, by virtue of which it tends to regain its original shape when the external force is removed, is Elasticity.

Reason (R) : The restoring force depends upon the bonded inter atomic and inter molecular force of solid.

In the light of the above statements, choose the **correct** answer from the options given below :

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**

(2) **(A)** is false but **(R)** is true

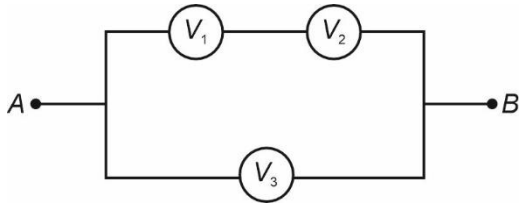
(3) **(A)** is true but **(R)** is false

(4) Both **(A)** and **(R)** are true but **(R)** is **not** the correct explanation of **(A)**

Answer (1)

Sol. The elasticity is responsible for shape regain of material and it cause due to inter atomic bond and intermolecular bond.

32. Three voltmeters, all having different internal resistances are joined as shown in figure. When some potential difference is applied across A and B, their readings are V_1, V_2 and V_3. Choose the correct option.



(1) $V_1 + V_2 > V_3$ (2) $V_1 = V_2$

(3) $V_1 \neq V_3 - V_2$ (4) $V_1 + V_2 = V_3$

Answer (4)

Sol. V_1 and V_2 are in series and they are in parallel with V_3.

$\therefore \quad V_1 + V_2 = V_3$

33. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Work done by electric field on moving a positive charge on an equipotential surface is always zero.

Reason (R) : Electric lines of forces are always perpendicular to equipotential surfaces.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) **(A)** is correct but **(R)** is not correct

(2) **(A)** is not correct but **(R)** is correct

(3) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**

(4) Both **(A)** and **(R)** are correct but **(R)** is **not** the correct explanation of **(A)**

Answer (3)

Sol. The potential at equipotential surface remain same everywhere, so no work is needed to move any charged particle and $\vec{E}$ is perpendicular to equipotential surface.



34. Primary side of a transformer is connected to 230 V, 50 Hz supply. Turns ratio of primary to secondary winding is 10 : 1. Load resistance connected to secondary side is 46 Ω. The power consumed in it is

(1) 10.0 W (2) 12.0 W

(3) 11.5 W (4) 12.5 W

Answer (3)

Sol. $\because \quad \frac{N_p}{N_s} = \frac{V_p}{V_s}$

$\Rightarrow \quad V_s = 23 \text{ V}$

$\Rightarrow \text{Power} = \frac{V_s^2}{R} = 11.5 \text{ W}$

35. A current of 200 μA deflects the coil of a moving coil galvanometer through 60°. The current to cause deflection through $\frac{\pi}{10}$ radian is

(1) 120 μA (2) 180 μA

(3) 30 μA (4) 60 μA

Answer (4)

Sol. Current $(i) = \frac{200\ \mu}{\left(\frac{\pi}{3}\right)} \times \frac{\pi}{10} = 60\ \mu\text{A}$

36. The threshold frequency of a metal with work function 6.63 eV is :

(1) 1.6×10^{15} Hz (2) 16×10^{12} Hz

(3) 1.6×10^{12} Hz (4) 16×10^{15} Hz

Answer (1)

Sol. Here $\phi = 6.63$ eV, $h = 6.63 \times 10^{-34}$ J.s

$$v = \frac{\phi}{h} = 1.6 \times 10^{15} \text{ Hz}$$

37. Given below are two statements :

Statement (I) : The limiting force of static friction depends on the area of contact and independent of material.

Statement (II) : The limiting force of kinetic friction is independent of the area of contact and depends on materials.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) **Statement I** is correct but **Statement II** is incorrect

(2) **Statement I** is incorrect but **Statement II** is correct

(3) Both **Statement I** and **Statement II** are incorrect

(4) Both **Statement I** and **Statement II** are correct

Answer (2)

Sol. Friction force is independent on area and it only depends on nature of material in contact.

38. A bullet is fired into a fixed target looses one third of its velocity after travelling 4 cm. It penetrates further $D \times 10^{-3}$ m before coming to rest. The value of D is

(1) 5

(2) 3

(3) 4

(4) 2

Answer (No option is correct)



Sol. $v = \frac{2u}{3}$

$v^2 - u^2 = 2a(0.04)$

$-\frac{5}{9}u^2 = 2a \times (0.04)$

$-\frac{u^2}{2a} = \frac{9}{5} \times 0.04$

$\therefore$ distance travelled further

$= \frac{9}{5} \times 0.04 - 0.04 = 32 \times 10^{-3}$

$D = 32$

None of the option is correct.

39. A ball suspended by a thread swings in a vertical plane so that its magnitude of acceleration in the extreme position and lowest position are equal. The angle (θ) of thread deflection in the extreme position will be :

(1) $2\tan^{-1}\left(\frac{1}{2}\right)$ (2) $\tan^{-1}\left(\sqrt{2}\right)$

(3) $\tan^{-1}\left(\frac{1}{2}\right)$ (4) $2\tan^{-1}\left(\frac{1}{\sqrt{5}}\right)$

Answer (1)

Sol. At extreme position

$a_1 = g\sin\theta$

At mean position

$a_2 = \frac{v^2}{\ell}$

$a_2 = 2g(1-\cos\theta)$

$2g(1-\cos\theta) = g\sin\theta$

$\tan\frac{\theta}{2} = \frac{1}{2}$

$\theta = 2\tan^{-1}\left(\frac{1}{2}\right)$

40. Wheatstone bridge principle is used to measure of specific resistance (S_1) of given wire, having length L, radius r. If X is the resistance of wire, then specific resistance is ; $S_1 = X\left(\frac{\pi r^2}{L}\right)$. If the length of the wire gets doubled, then the value of specific resistance will be

(1) S_1 (2) $\frac{S_1}{2}$

(3) $\frac{S_1}{4}$ (4) $2S_1$

Answer (1)

Sol. $S_1 = X\left(\frac{\pi r^2}{L}\right)$

Specific resistance, independent on the other factor.

So, it will remain same.

41. The total kinetic energy of 1 mole of oxygen at 27°C is

[Use universal gas constant (R) = 8.31 J/mole K]

(1) 6232.5 J (2) 5670.5 J

(3) 5942.0 J (4) 6845.5 J

Answer (1)

Sol. Kinetic energy = $\frac{5}{2}RT$

$= \frac{5}{2} \times 8.31 \times 300$

= 6232.5 J

42. A heavy iron bar of weight 12 kg is having its one end on the ground and the other on the shoulder of a man. The rod makes an angle 60° with the horizontal, the weight experienced by the man is :

(1) $6\sqrt{3}$ kg (2) 12 kg

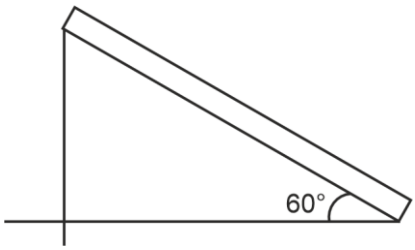
(3) 3 kg (4) 6 kg

Answer (4)



Sol. Net torque about point of contact on ground

$\tau_{net} = 0$



i.e., $N(l\cos 60) = mg \times \frac{l}{2}$

N = 60 newton

= 6 kg

43. During an adiabatic process, the pressure of a gas is found to be proportional to the cube of its absolute temperature. The ratio of $\frac{Cp}{Cv}$ for the gas is :

(1) $\frac{7}{5}$

(2) $\frac{5}{3}$

(3) $\frac{3}{2}$

(4) $\frac{9}{7}$

Answer (3)

Sol. Here, process is adiabatic

and $P \propto T^3$

$\Rightarrow PT^{-3} = \text{const}$

or $pV^{\frac{3}{2}} = \text{const}$

so $\frac{Cp}{Cv} = \frac{3}{2}$

44. Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : The angular speed of the moon in its orbit about the earth is more than the angular speed of the earth in its orbit about the sun.

Reason (R) : The moon takes less time to move around the earth than the time taken by the earth to move around the sun.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**

(2) **(A)** is correct but **(R)** is not correct

(3) **(A)** is not correct but **(R)** is correct

(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**

Answer (4)

Sol. Angular speed of moon (ω_m) > Angular speed of Earth (ω_e)

Time period $(T) = \frac{2\pi}{\omega}$

So moon take less time to move around the earth than earth around sun.

45. Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : In Vernier calliper if positive zero error exists, then while taking measurements, the reading taken will be more than the actual reading.

Reason (R) : The zero error in Vernier Calliper might have happened due to manufacturing defect or due to rough handling.



In the light of the above statements, choose the correct answer from the options given below :

(1) **(A)** is true but **(R)** is false

(2) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**

(3) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**

(4) **(A)** is false but **(R)** is true

Answer (3)

Sol. In case of positive error, measured value will be more than real and it may happen due to manufacturing defect or due to rough handling.

46. The equation of state of a real gas is given by $\left(P+\frac{a}{V^2}\right)(V-b)=RT$, where P, V and T are pressure, volume and temperature respectively and R is the universal gas constant. The dimensions of $\frac{a}{b^2}$ is similar to that of :

(1) P

(2) RT

(3) R

(4) PV

Answer (1)

Sol. $\left(P+\frac{a}{V^2}\right)(V-b)=RT$

$[a] = [RTV]$

$[b]=\left[\frac{RT}{P}\right]$

$[ab]=[RTV^2]$

$[PV] = [RT]$

Then $\left[\frac{a}{b^2}\right]=\left[\frac{RTV}{(RT)^2}P^2\right]=[P]$

47. The atomic mass of ${}_6C^{12}$ is 12.000000 u and that of ${}_6C^{13}$ is 13.003354 u. The required energy to remove a neutron from ${}_6C^{13}$, if mass of neutron is 1.008665 u, will be :

(1) 49.5 Mev

(2) 62.5 MeV

(3) 6.25 Mev

(4) 4.95 MeV

Answer (4)

Sol. ${}_6C^{13} \longrightarrow {}_6C^{12} + {}_0n^1$

$\Delta E = \Delta mc^2$

= (0.005311) × 931.5 MeV

= 4.95 MeV

48. An object is placed in a medium of refractive index 3. An electromagnetic wave of intensity 6×10^8 W/m^2 falls normally on the object and it is absorbed completely. The radiation pressure on the object would be (speed of light in free space $= 3 \times 10^8$ m/s):

(1) 6 Nm^{-2}

(2) 18 Nm^{-2}

(3) 2 Nm^{-2}

(4) 36 Nm^{-2}

Answer (1)

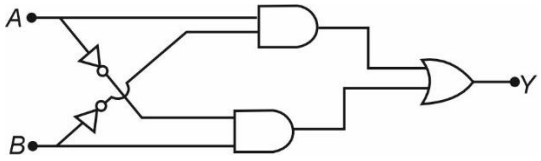
Sol. Radiation pressure $(P) = \frac{I_0}{C'}$

$$= \frac{6\times10^8}{\left(\frac{3\times10^8}{3}\right)}$$

= 6 Nm^{-2}



49. The truth table of the given circuit diagram is :



(1)

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

(2)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

(3)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

(4)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

Answer (2)

Sol. $Y = A \cdot \bar{B} + B \cdot \bar{A}$

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

50. When a polaroid sheet is rotated between two crossed polaroids then the transmitted light intensity will be maximum for a rotation of:

(1) 30°

(2) 45°

(3) 60°

(4) 90°

Answer (2)

Sol. $I = I_0 \cos^2\theta \cdot \sin^2\theta$

$$I = I_0\left(\frac{2\sin\theta \cdot \cos\theta}{2}\right)^2$$

$$= \frac{I_0}{4}(\sin 2\theta)^2$$

$I \rightarrow$ max, $\sin 2\theta = 1$

$\theta = 45°$

SECTION - B

Numerical Value Type Questions: This section contains 10 Numerical based questions. Attempt any 5 questions out of 10. The answer to each question should be rounded-off to the nearest integer.

51. A closed organ pipe 150 cm long gives 7 beats per second with an open organ pipe of length 350 cm, both vibrating in fundamental mode. The velocity of sound is _____ m/s.

Answer (294)

Sol. For close pipe: $n_1 = \frac{V}{4L_1} = \frac{V}{6}$

For open pipe: $n_2 = \frac{V}{2L_2} = \frac{V}{7}$

$n_1 - n_2 = 7$

$V = 294.00$ m/s



52. A body falling under gravity covers two points A and B separated by 80 m in 2 s. The distance of upper point A from the starting points is _____ m (use $g = 10\ ms^{-2}$).

Answer (45)

Sol. Let time taken to reach point A is t,

Then $V_A = gt$

$V_B = gt + 2g$

$80 = (gt)(2) + \frac{1}{2}9(2)^2$

$t = 3$ sec.

Distance of point A, $S = \frac{1}{2}(g)(3)^2 = 45$ m

53. Two charges of –4 μC and + 4 μC are placed at the point $A(1, 0, 4)$ m and $B(2, -1, 5)$ m located in an electric field $\vec{E} = 0.20\ \hat{i}$ V/cm. The magnitude of the torque acting on the dipole is $8\sqrt{\alpha} \times 10^{-5}$ Nm, where α = _____.

Answer (2)

Sol. Torque $(\vec{\tau}) = \vec{P} \times \vec{E}$

$= q(\vec{l} \times \vec{E})$

$= 4 \times 10^{-6}(\hat{i} - \hat{j} + \hat{k}) \times (0.2\hat{i}) \frac{V}{10^{-2}\,m}$

$|\vec{\tau}| = 8\sqrt{2} \times 10^{-5}\ N \cdot m$

54. The electric potential at the surface of an atomic nucleus (z = 50) of radius 9 × 10^{-13} cm is _____ × 10^6 V.

Answer (8)

Sol. Potential$(V) = \frac{Kz(e)}{r}$

$= \frac{9 \times 10^9 \times 50(1.6 \times 10^{-19})}{9 \times 10^{-15}}$

= 8 × 10^6 V

55. If Rydberg's constant is R, the longest wavelength of radiation in Paschen series will be $\frac{\alpha}{7R}$, where α = _____.

Answer (144)

Sol. $\frac{1}{\lambda} = RZ^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$

For given data, $n_1 = 3$

$n_2 = 4$

$Z_0 = 1$

$\frac{1}{\lambda} = R\left(\frac{1}{9} - \frac{1}{16}\right)$

$\frac{1}{\lambda} = \frac{7R}{144}$

$\lambda = \frac{144}{7R}$

56. A series LCR circuit with $L = \frac{100}{\pi}$ mH, $C = \frac{10^{-3}}{\pi}$ F and $R = 10\ \Omega$, is connected across an ac source of 220 V, 50 Hz supply. The power factor of the circuit would be _____ .

Answer (1)

Sol. Power factor = $\frac{R}{Z}$

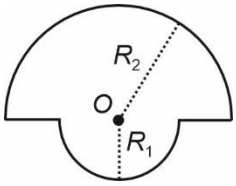
here $X_C = X_L = 10\ \Omega$

so Z = 10 Ω

power factor = $\frac{10}{10} = 1$



57. The magnetic field at the centre of a wire loop formed by two semicircular wires of radii $R_1 = 2\pi$ m and $R_2 = 4\pi$ m, carrying current $I = 4$ A as per figure given below is $\alpha \times 10^{-7}$ T. The value of α is _____ . (Centre O is common for all segments)



Answer (3)

Sol. Net magnetic field $(B) = B_1 + B_2$

$$= \frac{\mu_0 I}{4}\left(\frac{1}{R_1} + \frac{1}{R_2}\right)$$

$$= \frac{\mu_0 I}{4} \times \frac{3}{4\pi}$$

$$= 3 \times 10^{-7} \text{ T}$$

58. The reading of pressure metre attached with a closed pipe is 4.5×10^4 N/m^2. On opening the valve, water starts flowing and the reading of pressure metre falls to 2.0×10^4 N/m^2. The velocity of water is found to be $\sqrt{V}$ m/s. The value of V is _____ .

Answer (50)

Sol. Change in pressure $(\Delta P) = \frac{1}{2}\rho V^2$

$$2.5 \times 10^4 = \frac{1}{2} \times 10^3 V^2$$

$$V = \sqrt{50} \text{ m/s}$$

59. A ring and a solid sphere roll down the same inclined plane without slipping. They start from rest. The radii of both bodies are identical and the ratio of their kinetic energies is $\frac{7}{x}$, where x is _____ .

Answer (7)

Sol. If mass of ring and solid sphere will be same, then total kinetic energy of ring will be equal to total kinetic energy of solid sphere.

so $\frac{(\text{K.E.})_{\text{ring}}}{(\text{K.E.})_{\text{sphere}}} = 1$

60. A parallel beam of monochromatic light of wavelength 5000 Å is incident normally on a single narrow slit of width 0.001 mm. The light is focused by convex lens on screen, placed on its focal plane. The first minima will be formed for the angle of diffraction of _____ (degree).

Answer (29)

Sol. Angle of diffraction = $\frac{\lambda}{a}$ rad

$$= \frac{500 \times 10^{-10} \times 10}{10^{-6}} = \frac{10}{20} \text{ rad}$$

$$= 28.63°$$

$$\approx 29$$



CHEMISTRY

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

61. Identify from the following species in which d^2sp^3 hybridization is shown by central atom :

(1) BrF_5 (2) SF_6

(3) $[Pt(Cl_4)]^{2-}$ (4) $[Co(NH_3)_6]^{3+}$

Answer (4)

Sol. Co^{3+} : $3d^6\,4s^0$

Pairing will take place and hybridization will be d^2sp^3.

62. Which structure of protein remains intact after coagulation of egg white on boiling?

(1) Tertiary (2) Quaternary

(3) Secondary (4) Primary

Answer (4)

Sol. Primary structure of protein remains intact (it represents the sequence of amino acids).

63. Choose the correct option having all the elements with d^{10} electronic configuration from the following :

(1) ^{27}Co, ^{28}Ni, ^{26}Fe, ^{24}Cr

(2) ^{46}Pd, ^{28}Ni, ^{26}Fe, ^{24}Cr

(3) ^{29}Cu, ^{30}Zn, ^{48}Cd, ^{47}Ag

(4) ^{28}Ni, ^{24}Cr, ^{26}Fe, ^{29}Cu

Answer (3)

Sol. Cu : $4s^13d^{10}$

Zn : $4s^23d^{10}$

Cd : $5s^24d^{10}$

Ag : $5s^14d^{10}$

64. The final product A, formed in the following reaction sequence is

$$Ph-CH=CH_2 \xrightarrow[\text{(iii) HBr}]{\text{(i) } BH_3,\ \text{(ii) } H_2O_2,\ \ominus OH} \xrightarrow{\text{(iv) Mg, ether, then } HCHO/H_3O^+} A$$

(1) $Ph-CH_2-CH_2-CH_3$

(2) $Ph-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_3$

(3) $Ph-CH_2-CH_2-CH_2-OH$

(4) $Ph-\underset{\displaystyle CH_2OH}{\underset{|}{CH}}-CH_3$

Answer (3)

Sol.

$$Ph-CH=CH_2 \xrightarrow[(ii)\ H_2O_2/OH^-]{(i)\ BH_3} Ph-CH_2-\underset{\displaystyle OH}{\underset{|}{CH_2}}$$

$$Ph-CH_2-\underset{\displaystyle OH}{\underset{|}{CH_2}} \xrightarrow{HBr} Ph-CH_2-CH_2-Br$$

$$Ph-CH_2-CH_2-Br \xrightarrow[\text{then } H-\overset{\displaystyle O}{\overset{\|}{C}}-H\ /\ H_3O^+]{Mg,\ ether} Ph-CH_2-\underset{\displaystyle OH}{\underset{|}{CH_2}}-CH_2$$

(A)

Option (3) is correct.

